Roll No.: | | | | | | | | | | | | |

Test Date: 30-08-2026

Aakash
Medical | IIT-JEE | Foundations

C
Booklet Code

Aakash Delhi/NCR Zone

AIATS
ALL INDIA AAKASH TEST SERIES

for

Medical Entrance Exam - 2028

National Eligibility-cum-Entrance Test (NEET)

TEST No. 1

(XI Studying Students)

(Phase-1 & 2)

INSTRUCTIONS FOR CANDIDATES

1. Read each question carefully.
2. It is mandatory to use Blue/Black Ball Point Pen to darken the appropriate circle in the answer sheet.
3. Mark should be dark and should completely fill the circle.
4. Rough work must not be done on the answer sheet.
5. Do not use white-fluid or any other rubbing material on answer sheet. No change in the answer once marked is allowed.
6. Student cannot use log tables and calculators or any other material in the examination hall.
7. Before attempting the question paper, student should ensure that the test paper contains all pages and no page is missing.
8. Before handing over the answer sheet to the invigilator, candidate should check that Roll No. and Centre Code have been filled and marked correctly.
9. Immediately after the prescribed examination time is over, the answer sheet to be returned to the invigilator.
10. (i) The question paper consists of 180 compulsory questions (45 questions each in Physics and Chemistry and 90 questions in Biology (Botany & Zoology)). The total duration of the examination will be 180 minutes (3 Hrs). (ii) Each correct answer carries four marks. One mark will be deducted for each incorrect answer from the total score.

Note : It is compulsory to fill **Roll No.** and **Test Booklet Code** on answer sheet, otherwise your answer sheet will not be considered.

Test No. 1

TOPICS OF THE TEST

Physics	Units and Measurements, Motion in a Straight Line
Chemistry	Some Basic Concepts of Chemistry, Structure of Atom
Botany	Cell: The Unit of Life
Zoology	Structural Organisation in Animals: Animal Tissues only, Biomolecules

Aakash Medical IIT-JEE Foundations (AR-001)

MM : 720

TEST - 1

Time : 3 Hrs.

[PHYSICS]

Choose the correct answer:

1. Two particles of different masses are dropped from height of 16 m and 25 m respectively. The ratio of time taken by them to reach ground is

   (1) $\frac{16}{25}$ (2) $\frac{3}{5}$

   (3) $\frac{4}{5}$ (4) $\frac{4}{3}$

2. A tiny metallic rectangular sheet has a length and breadth of 5.00 mm and 2.50 mm respectively. Using a screw gauge with a pitch of 0.75 mm and 15 divisions on the circular scale, the maximum fractional error in the area of sheet will be

   (1) 0.3 (2) 0.03

   (3) 1.5 (4) 3

3. A ball is projected vertically upward from point P with an initial velocity of 40 m s^{-1} at time $t = 0$. At $t = 2$ s another ball is projected vertically upward with same velocity from point P. Second ball will meet the first ball at t equal to

   (1) 4 s (2) 8 s

   (3) 6 s (4) 5 s

4. A car A is standing 100 m behind another car B, which is also at rest. The two cars start moving in the same direction at the same instant but with different forward acceleration. If car B has acceleration 1 m/s^2 and car A has acceleration 3 m/s^2, then the car A will catch up with the car B after a time

   (1) 10 s (2) 20 s

   (3) 5 s (4) 33.3 s

5. A ball is thrown vertically upward from a moving elevator with an initial speed of 10 m/s relative to the elevator. If elevator is moving up with speed 5 m/s, then initial speed of ball relative to the ground is

   (1) 10 m/s (2) 5 m/s

   (3) 15 m/s (4) $5\sqrt{5}$ m/s

6. A train 100 m long moving at 30 m/s crosses another train 150 m long moving at 20 m/s in the opposite direction. The time taken by both of them to completely pass each other is

   (1) 12.5 s (2) 5 s

   (3) 10 s (4) 25 s

7. A particle is moving along the x-axis with its position varying with time t as $x = a(t^2 - 2) + b(t - 2)^2$, where a and b are constants. The correct option is

   (1) The acceleration of the particle is $2b$

   (2) The acceleration of the particle is $2a$

   (3) The initial velocity of the particle is $2b$

   (4) The acceleration of the particle is $2(a + b)$

8. A man travelling on a straight line moves with a uniform velocity $u = 5$ m/s for a distance d and with a uniform velocity V_0 for the next $1.5d$ distance. If the average velocity is $\frac{50}{7}$ m/s, then V_0 is

   (1) 12 m/s

   (2) 8 m/s

   (3) 10 m/s

   (4) 15 m/s

Space for Rough Work

9. A ball is dropped from the top of a tower of height 60 m. At a height h from the ground, the magnitude of its velocity becomes exactly equal to twice the magnitude of acceleration due to gravity. The value of h is [take $g = 10\ m/s^2$]
   (1) 20 m (2) 30 m
   (3) 40 m (4) 10 m

10. The pair of physical quantities not having the same dimensions, is
   (1) Pressure and Young's modulus
   (2) Torque and energy
   (3) Surface tension and Impulse
   (4) Angular momentum and Planck's constant

11. A body takes 10 s to reach its maximum speed of 90 km/h from rest. The magnitude of its average acceleration is
   (1) $2.5\ m/s^2$ (2) $5\ m/s^2$
   (3) $4.5\ m/s^2$ (4) $9\ m/s^2$

12. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
   Assertion (A): The zero velocity of a particle at any instant does not necessarily imply zero acceleration at that instant.
   Reason (R): Acceleration must result due to change in speed only.
   In the light of above statements, select the correct option.
   (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
   (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
   (3) (A) is true but (R) is false
   (4) Both (A) and (R) are false

13. The position of a particle moving on x-axis is given by $x(t) = A\sin t + B\cos^2 t + Ct^2 + D$, where t is time while A, B, C and D are constants. The dimensions of $\frac{ABC}{D}$ is
   (1) $[M^0L^3T^{-2}]$ (2) $[M^0L^2T^0]$
   (3) $[M^0L^2T^{-2}]$ (4) $[M^0LT^0]$

14. A physical quantity Z is related to four other quantities p, q, r and s as $Z = \frac{pq^2}{r^3\sqrt{s}}$. If the percentage errors in p, q, r and s are 1%, 2%, 3% and 2% respectively, then maximum percentage error in Z will be
   (1) 14% (2) 10%
   (3) 15% (4) 18%

15. Balls are dropped vertically downward from the top of a tower at uniform time interval of T. If at the instant the 5th ball is dropped, the separation between the 2nd and 3rd ball is found to be 35 m, then the time interval T will be
   (1) $\sqrt{\frac{7}{3}}$ s (2) $\frac{\sqrt{7}}{3}$ s
   (3) $\sqrt{\frac{7}{5}}$ s (4) $\frac{\sqrt{7}}{5}$ s

16. In a hypothetical system of units, the fundamental quantities are chosen such that the unit of mass is α kg, the unit of length is β m, and the unit of time is γ s. If a force of 10 N acts on a body, then its numerical value in this new system is
   (1) $10\alpha\beta\gamma^{-2}$
   (2) $10\alpha^{-1}\beta^{-1}\gamma^2$
   (3) $10\alpha\beta^{-1}\gamma^2$
   (4) $10\alpha^{-1}\beta\gamma^{-2}$

Space for Rough Work

17. Final result of addition of three measurements 0.1 cm, 0.01 cm and 0.001 cm will be

(1) Zero (2) 0.111 cm

(3) 0.11 cm (4) 0.1 cm

18. The least count of a vernier calliper is 0.01 cm and it has a zero error of (–0.04) cm. While measuring the diameter of a cylinder, the main scale reading is 2.4 cm and the 6th vernier scale division coincides with a main scale division. The correct diameter of the cylinder will be

(1) 250 mm (2) 2.5 mm

(3) 0.25 mm (4) 25 mm

19. In an experiment to estimate acceleration due to gravity (g) using a simple pendulum, the length of the pendulum is measured as 40.0 cm using a meter scale with a least count of 1 mm. The total time taken for 40 complete oscillations is 80 s using a stopwatch with a resolution of 0.5 s. The maximum percentage error in the determination of g is [Time period of simple pendulum, $T = 2\pi\sqrt{\frac{l}{g}}$]

(1) 3% (2) 1.5%

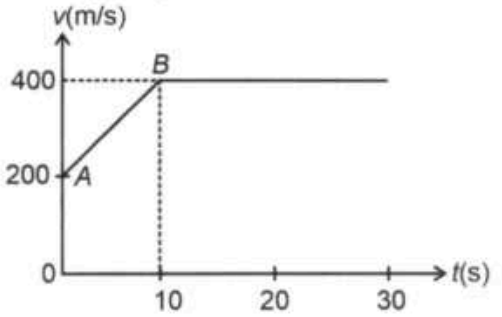
(3) 2% (4) 2.5%

20. A particle moves along a straight line such that its acceleration is given as a function of velocity $a = -K\sqrt{v}$, where K is a positive constant. If the initial velocity of the particle at time $t = 0$ is u, then the total time taken by the particle to come to rest, is

(1) $2\sqrt{\frac{u}{K}}$ (2) $\frac{2\sqrt{u}}{K}$

(3) $\frac{\sqrt{u}}{K}$ (4) $\sqrt{\frac{u}{K}}$

21. The motion of a particle is described by the velocity-time ($v - t$) graph as shown below. The distance covered by the particle in the first 25 seconds is



(1) 12 km (2) 9 km

(3) 11 km (4) 11.6 km

22. The dimensional formula for change in energy is

(1) $[ML^2T^{-2}]$ (2) $[MLT^{-2}]$

(3) $[ML^{-1}T^{-1}]$ (4) $[ML^2T^{-1}]$

23. The least count of a screw gauge is 0.01 mm. If pitch of instrument is increased by 75% and the number of divisions on the circular scale is reduced by 50%, the new least count will be

(1) 15×10^{-3} mm (2) 20×10^{-3} mm

(3) 25×10^{-3} mm (4) 35×10^{-3} mm

24. A particle moves along a straight line with an acceleration $a = (6t + 2)$ m/s^2. If the particle starts from rest at time $t = 0$, the displacement of the particle after 2 seconds will be

(1) 12 m (2) 16 m

(3) 8 m (4) 14 m

25. A vehicle travels half the distance with speed u and remaining distance with speed $3u$. Its average speed is

(1) $\frac{2u}{3}$ (2) $\frac{4u}{3}$

(3) $\frac{3u}{4}$ (4) $\frac{3u}{2}$

Space for Rough Work

26. If $y = 3t^2 + 2t$, then value of $\frac{dy}{dt}$ at $t = 1$ s is

(1) 3 (2) 5

(3) 8 (4) 6

27. If the measurements of mass $m = (0.60 \pm 0.003)$ g, radius $r = (0.50 \pm 0.01)$ cm and length $l = (10.00 \pm 0.05)$ cm are given for a cylindrical wire, then the maximum percentage error in the measurement of its density will be (in %)

(1) 5 (2) 7

(3) 4 (4) 8

28. The velocity (v) of a particle moving along a straight line depends on time (t) as $v = \alpha t - \beta t^2$, where α and β are positive constants. The displacement of the particle between the two moments when it is momentarily at rest is

(1) $\frac{\beta^3}{6\alpha^2}$ (2) $\frac{2\beta^3}{\alpha^2}$

(3) $\frac{3\alpha^3}{2\beta^2}$ (4) $\frac{\alpha^3}{6\beta^2}$

29. The velocity-position ($v - s$) relation of a particle moving in a straight line is given by $v^2 = s^2 + 2s + 4$. (v is in m s^{-1} and s is in m)

The acceleration of the particle when $s = 3$ m, will be

(1) 4 m/s^2 (2) 7 m/s^2

(3) 10 m/s^2 (4) 1 m/s^2

30. The correct statements for an object projected vertically upwards from the ground until it reaches its maximum height are (neglect air resistance):

(A) The acceleration of the object is uniform

(B) The speed decreases linearly with time

(C) Distance covered in each second increases

Choose the correct option among the following.

(1) Only (A) and (C) are correct

(2) Only (B) and (C) are correct

(3) Only (A) and (B) are correct

(4) All (A), (B) and (C) are correct

31. A ball is projected vertically upward from the ground with some initial velocity. If T_1 is the time taken to reach first half of its maximum height and T_2 is the time taken to travel second half of the maximum height respectively, then choose the correct relation between T_1 and T_2.

(1) $T_2 = (\sqrt{2} - 1)T_1$ (2) $T_2 = \sqrt{2}T_1$

(3) $T_1 = (\sqrt{2} - 1)T_2$ (4) $T_1 = (\sqrt{2} + 1)T_2$

32. A particle starts from rest and accelerates uniformly with 1 m/s^2 for 5 s and then retards uniformly with same rate till it comes to rest. The distance covered by particle is

(1) 12 m (2) 12.5 m

(3) 30 m (4) 25 m

33. A hot air balloon starts moving upwards from the ground with a constant acceleration of 2 m/s^2. After 10 seconds of its journey, a ball is dropped from it. The maximum height reached by the ball from the ground is

(1) 120 m (2) 100 m

(3) 20 m (4) 40 m

34. A bullet is fired horizontally into a uniform wooden block with an initial velocity V_0. After penetrating 36 cm into the block, its velocity is reduced to $\frac{V_0}{2}$.

Then how much further will the bullet penetrate before coming to rest?

(1) 36 cm (2) 48 cm

(3) 18 cm (4) 12 cm

Space for Rough Work

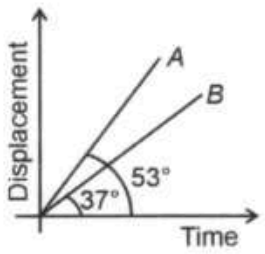
35. A ball is projected vertically upward with velocity 20 m/s and it returns to the point of projection on the ground. For the complete journey (Take, $g = 10\ m/s^2$)
    (1) Its average speed is 10 m/s
    (2) Its average speed is 20 m/s
    (3) Its average acceleration is $10\ m/s^2$
    (4) Both (1) and (3) are correct

36. The equation for real gas is given by $\left(P+\frac{a}{V^2}\right)(V-b)=RT$, where symbols have their usual meaning. The dimensions of $\frac{a}{b^2}$ is equivalent to that of
    (1) Energy density (2) Planck's constant
    (3) Strain (4) Compressibility

37. The number of insignificant zeros in 0.006200 is
    (1) Five (2) Three
    (3) Two (4) Four

38. Consider the following statements:
    Statement (A): For uniform acceleration, velocity versus displacement graph will not be a straight line.
    Statement (B): Both acceleration and velocity can change values within a time interval.
    The correct statement(s) is/are
    (1) Only statement (A)
    (2) Only statement (B)
    (3) Both (A) and (B)
    (4) Neither (A) nor (B)

39. The displacement time graphs of two moving particles *A* and *B* make an angle of 37° and 53° with the time axis as shown in figure. The ratio of their velocity is



    (1) $\frac{16}{9}$ (2) $\frac{3}{5}$
    (3) $\frac{4}{5}$ (4) $\frac{7}{9}$

40. Which of the following are the sources of systematic errors?
    (1) Instrumental errors
    (2) Imperfection in experimental technique or procedure
    (3) Personal error
    (4) All of these

41. The velocity of a car changes with time according to the equation, $v = 10 - t^3$. For time ($0 < t < 2$ s), the car is undergoing
    (1) Uniform acceleration
    (2) Non-uniform acceleration
    (3) Uniform retardation
    (4) Non-uniform retardation

42. The given equation shows the variation of velocity (v) with time (t), $v = At^2 + \frac{Bt}{C+t}$, where *A*, *B* and *C* are constants. The dimensions of *ABC* will be
    (1) $[M^0LT^{-2}]$
    (2) $[M^0LT^{-3}]$
    (3) $[M^0L^2T^{-3}]$
    (4) $[M^0L^2T^{-2}]$

Space for Rough Work

43. Which among the following is a fundamental physical quantity?

(1) Electric current (2) Plane angle

(3) Force (4) Both (1) and (2)

44. Match the graphical interpretations of kinematics quantities in column-I with their corresponding physical meanings in column-II and choose the correct option.

	Column-I		Column-II
a.	Area under velocity time curve projected on time axis	(i)	Acceleration
b.	Area under acceleration time curve projected on time axis	(ii)	Change in velocity
c.	Slope of displacement time curve	(iii)	Displacement
d.	Slope of velocity time curve	(iv)	Velocity

(1) a(iii), b(ii), c(iv), d(i) (2) a(i), b(ii), c(iv), d(iii)

(3) a(iii), b(iv), c(ii), d(i) (4) a(ii), b(iii), c(iv), d(i)

45. A driver travelling at 30 m/s sees a red light. The driver has a reaction time of 0.5 s, after which he applies brakes, producing a constant deceleration of 5 m/s^2. The total stopping distance of the car from the moment, the driver first sees the red light, is

(1) 100 m (2) 90 m

(3) 105 m (4) 15 m

[CHEMISTRY]

46. Match List-I with List-II

	List-I (Sample)		List-II (Number of atoms)
a.	32 g of $CH_4(g)$	(i)	$6N_A$
b.	3.011×10^{23} molecules of SO_2 (g)	(ii)	$4N_A$
c.	44.8 L of CO_2 (g) at STP	(iii)	$10N_A$
d.	17 g of NH_3 (g)	(iv)	$1.5N_A$

The correct match is

(1) a(i), b(iii), c(ii), d(iv) (2) a(iii), b(iv), c(i), d(ii)

(3) a(ii), b(iii), c(i), d(iv) (4) a(iii), b(i), c(ii), d(iv)

47. If 28 g of nitrogen gas reacts with 8 g of hydrogen gas, then the maximum mass of ammonia gas produced in the reaction is

(1) 68 g

(2) 8.5 g

(3) 17 g

(4) 34 g

48. Calcium carbonate reacts with aqueous HCl according to the following reaction;

$CaCO_3(s)+2HCl(aq) \rightarrow CaCl_2(aq)+CO_2(g)+H_2O(l)$

The mass of $CaCO_3$ required to react completely with 50 mL of 0.5 M HCl is

(1) 1.25 g (2) 2.25 g

(3) 7.45 g (4) 3.45 g

Space for Rough Work

49. Consider the following statements:

(a) Law of multiple proportions is applicable to the pair of oxides SO_2 and SO_3.

(b) $Ni_{0.94}O$ follows law of definite proportions.

(c) Number of nucleons in 1.6 g of CH_4 is 1.6 N_A.

(d) A mixture of 4.6 g formic acid and 9.0 g oxalic acid ($H_2C_2O_4$) is treated with excess conc. H_2SO_4, the weight of evolved gases containing carbon atom(s) is 10.0 g.

(e) Significant figures in 200 eggs and 1.60 are same.

The correct statements are

(1) (a), (b), (c) and (e) only

(2) (a), (c) and (d) only

(3) (b) and (d) only

(4) (b), (c) and (e) only

50. If the mass percentage of zinc in an enzyme is 0.5%, then the minimum molar mass of the enzyme will be (Molar mass of zinc = 65 g mol^{-1})

(1) 26000 g mol^{-1} (2) 13000 g mol^{-1}

(3) 65000 g mol^{-1} (4) 29000 g mol^{-1}

51. Boron has two isotopes with atomic masses 10 u and 11 u. If its average atomic mass is 10.81 u, then the abundance of heavier isotope will be

(1) 19%

(2) 17%

(3) 81%

(4) 73%

52. Match List-I with List-II

	List-I (Multiple)		List-II (Prefix)
a.	10^{-18}	(i)	pico
b.	10^{-12}	(ii)	exa
c.	10^{12}	(iii)	atto
d.	10^{18}	(iv)	tera

The correct match is

(1) a(iii), b(i), c(ii), d(iv)

(2) a(i), b(iii), c(ii), d(iv)

(3) a(i), b(iii), c(iv), d(ii)

(4) a(iii), b(i), c(iv), d(ii)

53. Which of the following statements is incorrect according to Dalton's Atomic Theory?

(1) Matter consists of indivisible atoms.

(2) All the atoms of a given element have identical properties, including identical mass.

(3) Atoms can be created as well as destroyed in a chemical reaction.

(4) Compounds are formed when atoms of different elements combine in a fixed ratio.

54. 15 g of an impure sample of limestone liberates 4.4 g of CO_2 on strong heating.

The percentage purity of $CaCO_3$ in the sample of limestone is

(1) 33.23%

(2) 46.25%

(3) 66.67%

(4) 80.37%

55. $P + Q + 3R \rightarrow PQR_3$

Reaction of 4.0 g of P, 6.0×10^{23} atoms of Q, and 0.06 mol of R yields 9.4 g of compound, PQR_3. If the atomic mass of P and R are 40 u and 80 u, respectively, then atomic mass of Q is (Avogadro number = 6×10^{23})

(1) 140 u (2) 190 u

(3) 100 u (4) 110 u

56. If 0.1 mole of $BaCl_2$ is mixed with 0.2 mol of Na_3PO_4, then the maximum number of moles of $Ba_3(PO_4)_2$ that can be formed is

(1) 0.033 (2) 0.055

(3) 0.088 (4) 0.077

Space for Rough Work

57. Given below are two statements

Statement I: Mole fraction of the solute in 2 m aqueous solution is approximately 0.035.

Statement II: Both mole fraction and molality are temperature independent concentration terms.

In the light of above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

58. During the Ostwald's process, if 17 g of NH_3 (g) reacts with 44.8 L of O_2(g) at STP, then consider the following statements.

$4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(l)$

(a) NH_3 is limiting reagent
(b) 27 g of H_2O is formed
(c) 30 g of NO is formed

The correct statement(s) is/are

(1) (a) only (2) (a), (b) and (c)
(3) (b) and (c) only (4) (c) only

59. Given below are two statements.

Statement I: If 3.01×10^{22} molecules of glucose are present in 200 mL of its solution, then molarity of the solution will be 0.5 M.

Statement II: If 6.02×10^{23} molecules of glucose are present in 500 g water, then molality of the solution will be 2 m.

In the light of above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

60. How many grams of concentrated nitric acid solution should be used to prepare 500 mL of 1 M HNO_3? (The concentrated acid is 70% (w/w) HNO_3.)

(1) 45.0 g conc. HNO_3
(2) 90.0 g conc. HNO_3
(3) 60.0 g conc. HNO_3
(4) 30.0 g conc. HNO_3

61. Given below are two statements.

Statement I: If a mixture of gases contains O_2 and N_2 gases in the ratio of 2 : 3 (w/w) respectively, then the molar ratio of the two gases respectively in the mixture will be 7 : 12.

Statement II: Both 1 g of O_2 gas and 1 g of O_3 gases contain same number of oxygen atoms.

In the light of above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

62. An organic compound containing C, H and N gave the following analysis; C = 60%, H = 12% N = 28%. Its empirical formula would be

(1) $C_5H_{12}N_2$
(2) C_4H_6N
(3) $C_2H_3N_2$
(4) C_6H_5N

63. If 50.0 mL of 0.05 M HCl is mixed with 40.0 mL of 0.02 M $Ba(OH)_2$, then the concentration of H^+ ion, $[H^+]$ in the final solution will be

(1) 0.05 M
(2) 0.01 M
(3) 0.02 M
(4) 0.04 M

Space for Rough Work

64. Given below are two statements

Statement I: If 22 g of propane is completely reacted with 160 g of oxygen, then the mole fraction of CO_2 in the resulting mixture is 0.25.

Statement II: If a metal oxide contains 32% oxygen by mass, then equivalent weight of metal will be 17 g.

In the light of above statements, choose the correct answer from the options given below

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

65. A sample of a hydrate of barium chloride weighing 122 g was heated until all the water of hydration is removed. The dried sample weighed 104 g. The formula of the hydrated salt is (Atomic mass, Ba = 137 u, Cl = 35.5 u)

(1) $BaCl_2 \cdot H_2O$ (2) $BaCl_2 \cdot 3H_2O$

(3) $BaCl_2 \cdot 4H_2O$ (4) $BaCl_2 \cdot 2H_2O$

66. Given below are two statements: one is labelled as Assertion (A) and other is labelled as Reason (R)

Assertion (A): The energy of 2s orbital of hydrogen atom is greater than that of 2s orbital of lithium atom.

Reason (R): Energies of the orbitals in the same subshell decrease with increase in the atomic number.

In the light of above statements, choose the correct answer from the options given below

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

67. Match List-I with List-II

	List-I (State)		**List-II [Energy (in eV/atom)]**
a.	3rd excited state of He^+ ion	(i)	–13.6
b.	4th Bohr orbit of Be^{3+} ion	(ii)	–1.51
c.	2nd excited state of H atom	(iii)	–3.4
d.	4th Bohr orbit of H atom	(iv)	–0.85

The correct match is

(1) a(i), b(iii), c(iv), d(ii)

(2) a(iii), b(i), c(ii), d(iv)

(3) a(i), b(iii), c(ii), d(iv)

(4) a(iii), b(i), c(iv), d(ii)

68. Consider the following statements:

(a) Magnetic orbital quantum number, m_l, designates the orientation of the orbital.

(b) The total number of orbitals associated with the principal quantum number 4 is 16.

(c) The degeneracy of the third excited state of H atom is 9.

(d) Orbit angular momentum of electron in 2nd orbit is $2h$.

The correct statements are

(1) (a), (b) and (d) only

(2) (b) and (c) only

(3) (a), (b) and (c) only

(4) (a), (b), (c) and (d)

Space for Rough Work

69. Given below are two statements about photoelectric effect:

Statement I: The number of electrons ejected is proportional to the intensity of incident light.

Statement II: The kinetic energy of the ejected electrons increase with increase of frequency of the light used.

In the light of above statements, choose the correct answer from the options given below:

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

70. The ratio of the wavelength of the light absorbed by a hydrogen atom when it undergoes $n = 3 \rightarrow n = 4$ and $n = 4 \rightarrow n = 5$ transitions, respectively is

(1) 81 : 175 (2) 175 : 81

(3) 83 : 191 (4) 191 : 83

71. The orbital angular momentum of an electron in p orbital is

(1) $\frac{h}{2}$ (2) $\sqrt{2}h$

(3) $\sqrt{3}h$ (4) $\sqrt{6}h$

72. Match List-I with List II

	List-I Orbital		**List-II Radial node(s)**
a.	4*p*	(i)	1
b.	4*d*	(ii)	0
c.	5*s*	(iii)	2
d.	4*f*	(iv)	4

The correct match is

(1) a(iii), b(i), c(iv), d(ii) (2) a(i), b(iii), c(iv), d(ii)

(3) a(iii), b(i), c(ii), d(iv) (4) a(i), b(iii), c(ii), d(iv)

73. The de Broglie wavelength of an electron of H-atom in the 4th Bohr orbit is (a_0 = Bohr radius)

(1) $8\pi a_0$ (2) $4\pi a_0$

(3) $6\pi a_0$ (4) $2\pi a_0$

74. Consider the following sets of quantum numbers and choose the incorrect set of quantum number

	n	l	m_l	m_s
(1)	3	2	–1	$+\frac{1}{2}$
(2)	4	3	+4	$-\frac{1}{2}$
(3)	5	4	–3	$+\frac{1}{2}$
(4)	2	1	0	$-\frac{1}{2}$

75. Consider the following statements

(a) The phenomena black body radiation and photoelectric effect can be explained by particle nature of electromagnetic radiation.

(b) Work function of Li is higher than that of Mg.

(c) The maximum number of electrons in 3rd shell is 18.

(d) Balmer and Paschen spectral lines for the hydrogen spectrum lie in infrared region.

(e) The probability of finding an electron at a point within an atom is proportional to the square of the orbital wave function, $|\psi|^2$.

The correct statements are

(1) (a), (b) and (c) only

(2) (b), (d) and (e) only

(3) (a), (b), (c), (d) and (e)

(4) (a), (c) and (e) only

Space for Rough Work

76. Which of the following statements is incorrect?
(1) Two electrons occupying the same orbital are distinguished by spin quantum number
(2) Half-filled and fully-filled orbitals have greater stability due to greater exchange energy, greater symmetry and more balanced arrangement
(3) An orbital is designated by three quantum numbers while an electron in an atom is designated by four quantum numbers
(4) Both d_{xz} and d_{z^2} orbitals have electron density along the axes

77. The correct order of energies of atomic orbitals of hydrogen atom is
(1) $1s < 2s < 2p < 3s < 3p < 4s < 3d$
(2) $1s < 2s = 2p < 3s = 3p = 3d < 4s$
(3) $1s < 2s < 2p < 3s < 3p < 3d < 4s$
(4) $4s < 3d < 3p < 3s < 2p < 2s < 1s$

78. The de Broglie wavelength of a ball with mass 10 mg moving with a velocity of 100 ms^{-1} is
(1) 6.626×10^{-28} m (2) 6.626×10^{-33} m
(3) 6.626×10^{-31} m (4) 6.626×10^{-30} m

79. For multi-electron system, among the given atomic orbitals, which has highest energy?
(1) 5*d* (2) 5*f*
(3) 7*s* (4) 6*p*

80. The correct order of wavelength of following electromagnetic radiations is
(1) γ rays > X rays > UV > IR
(2) IR > UV > γ rays > X rays
(3) UV > IR > γ rays > X rays
(4) IR > UV > X rays > γ rays

81. When electromagnetic radiation of wavelength 300 nm falls on the surface of sodium, then electrons are emitted with a kinetic energy of 1.74×10^5 J mol^{-1}. What is the minimum energy needed to remove one mole of electrons from sodium? (Given: $h = 6.6 \times 10^{-34}$ Js, $N_A = 6 \times 10^{23}$)
(1) 2.2×10^4 J (2) 2.2×10^{-4} J
(3) 2.2×10^5 J (4) 2.2×10^{-5} J

82. The representation which violates Hund's rule of maximum multiplicity is
(1)  (2) 
(3)  (4) 

83. The total number of electrons in Cu atom having (n + l) value 4, is
(1) 8 (2) 6
(3) 7 (4) 10

84. Which one of the following represents all isoelectronic species?
(1) N_2O, NO_2, NO^+ (2) Mg^{2+}, Cl^-, O^{2-}
(3) Na^+, O^{2-}, Cl^- (4) NO^+, CO, CN^-

85. According to the Bohr theory, which of the following transitions in the hydrogen atom will give rise to the least energetic photon?
(1) n = 5 to n = 4 (2) n = 6 to n = 4
(3) n = 5 to n = 3 (4) n = 6 to n = 1

86. Consider the following statements
(a) If uncertainty in position is double of uncertainty of momentum, then uncertainty of velocity will be $\frac{1}{2m}\sqrt{\frac{h}{2\pi}}$.
(b) If the radius of third Bohr orbit of the Li^{2+} ion is x pm, then the radius of fourth Bohr orbit of Be^{3+} ion will be $\frac{4x}{3}$.
(c) Both Fe^{3+} and Cr^{2+} ions have same number of unpaired electrons.
The correct statement(s) is/are
(1) (a), (b) and (c) (2) (a) and (c) only
(3) (c) only (4) (a) and (b) only

Space for Rough Work

87. The total number of spectral lines obtained when electrons jump from 5th excited state to 1st excited state in hydrogen spectrum is
(1) 4 (2) 10
(3) 8 (4) 15

88. If wavelength of the first line of Lyman series of hydrogen atom is x nm, then the wavelength of the second line of Balmer series for hydrogen atom (in nm) will be
(1) 4x (2) 3x
(3) 5x (4) 7x

89. An ion 'X' with mass number 56 contains 3 units of positive charge and 30.4% more neutrons than electrons. The correct representation of ion 'X' is
(1) $^{56}_{26}X^{3+}$ (2) $^{56}_{30}X^{3+}$
(3) $^{56}_{25}X^{3+}$ (4) $^{56}_{27}X^{3+}$

90. Which of the following statements is incorrect about cathode rays?
(1) In the absence of electrical or magnetic field, these rays travel in straight line.
(2) These rays consist of positively charged particles, called protons.
(3) The characteristics of these rays do not depend upon the material of electrodes and the nature of the gas present in the cathode ray tube.
(4) They start from cathode and move towards the anode.

[BOTANY]

91. Match **column-I** with **column-II**.

	Column-I		Column-II
a.	RBCs	(i)	Double membrane bound structure that divide by fission
b.	Mitochondria	(ii)	Are an exception to cell theory
c.	*Mycoplasma*	(iii)	Contain approximately 52 percent proteins in the membrane
d.	Viruses	(iv)	Lacks non-living rigid structure that forms outer covering of cell

Choose the **correct** option.
(1) a(i), b(iii), c(iv), d(ii) (2) a(iv), b(iii), c(i), d(ii)
(3) a(iii), b(i), c(iv), d(ii) (4) a(ii), b(iv), c(i), d(iii)

92. Below are descriptions along with year of discoveries of certain cell structures (A-D). Identify the structures and select the correct option.
(A) It was first observed in 1898 near the [redacted]
(B) Its improved model was proposed by the scientists in 1972
(C) First observed under electron microscope as dense particles in 1953
(D) First described by the scientist as a cell organelle as early as [redacted]

	A	B	C	D
(1)	Endoplasmic reticulum	Chloroplast	Golgi apparatus	Mitochondria
(2)	Golgi apparatus	Plasma membrane	Ribosomes	Nucleus
(3)	Ribosomes	DNA	Golgi apparatus	Microbodies
(4)	Lysosomes	Plasma membrane	Golgi apparatus	Plastids

Space for Rough Work

93. Match **column-I** with **column-II**.

	Column-I		Column-II
a.	[redacted]	(i)	Do not play role in motility
b.	Mesosome	(ii)	Found in single celled organisms having false feet
c.	[redacted]	(iii)	Protect the cells and help in adhesion
d.	Contractile vacuole	(iv)	Help in cell wall formation

Choose the **correct** option.

(1) a(ii), b(iv), c(iii), d(i) (2) a(iv), b(ii), c(iii), d(i)

(3) a(iii), b(ii), c(iv), d(i) (4) a(iii), b(iv), c(i), d(ii)

94. Read the given statements of Assertion (A) and Reason (R) and choose the **correct** option.

Assertion (A): During different stages of cell division, cells show highly extended and elaborate nucleoprotein fibres in place of nucleus.

Reason (R): Chromatin is a loose, distinct elaborated network of DNA and some basic proteins in cytoplasm.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

95. Select the **correctly** matched pair.

(1)	Nucleus	–	Absent in mature sieve tubes and white blood cells
(2)	Plastids	–	Absent in tracheids and euglenoids
(3)	Cilia	–	Shows oar-like movement in BGA and PPLO
(4)	Polysome	–	Found in prokaryotes and synthesises proteins

96. Read the given statements and state them as true **(T)** or false **(F)**.

(A) Dictyosomes are found in [redacted]

(B) In chromoplasts, fat soluble carotenoid pigments like carotene, xanthophylls and anthocyanins are found

(C) The cisternae of Golgi apparatus closely resembles with [redacted]

(D) In eukaryotes, the shape of the cell may vary with the function they perform

(E) Microtubules mainly help in pseudopodia formation and are found in cytoskeleton of flagella

Choose the **correct** option.

	(A)	(B)	(C)	(D)	(E)
(1)	T	F	T	F	T
(2)	F	T	T	F	F
(3)	T	F	F	T	F
(4)	F	F	T	F	F

97. The scientist, who modified the cell theory in 1855 was

(1) British botanist and concluded, the presence of cell wall is a unique character of plant cells

(2) The first one to explain *Omnis cellula-e cellula*

(3) The first one to saw and describe a live cell under electron microscope

(4) Not able to explain how new cells were formed.

98. Which of the following can be found in both animal and higher plant cells?

(1) Histones (2) Centrioles

(3) Large sap vacuoles (4) Plastids

99. [redacted]

(a) Appear L-shaped during anaphase

(b) Are isobrachial

(c) Exhibit characteristic p and q arm

(d) Have terminal centromere

Choose the **correct** one(s):

(1) (a) and (c) only (2) (c) and (d) only

(3) (c) only (4) All except (d)

Space for Rough Work

100. Which of the given is [redacted] for primary cell wall of plants?
(1) Single layered
(2) Elastic in nature
(3) Capable of expansion in growing cells
(4) Form by deposition of wall material towards inner side of plasma membrane

101. Read the given statements and choose the **correct** option.

Statement (A): Free ribosomes synthesise structural proteins that remain within cells whereas the attached ribosomes synthesise proteins for transport.

Statement (B): The fluid nature of membrane is important from the point of view of functions like formation of intercellular junctions and endocytosis.

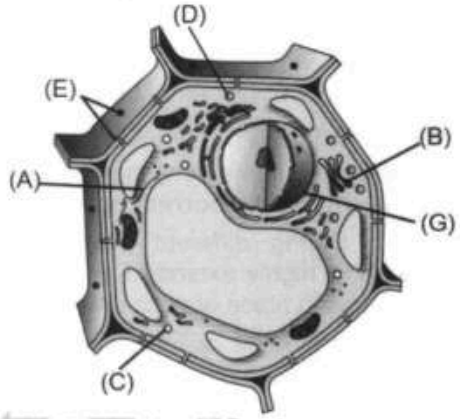
(1) Both statements (A) and (B) are true
(2) Both statements (A) and (B) are false
(3) Only statement (A) is true
(4) Only statement (B) is true

102. Select the **correctly** matched pair.

(1)	Primary lysosomes	–	Contain [redacted] hydrolytic enzymes
(2)	SER	–	Involved in detoxification of drugs
(3)	Bacterial cell	–	Have [redacted] complex but [redacted] simple cell envelope
(4)	Polysome	–	Single ribosome may attach to a multiple mRNA

103. Select the **odd** one out w.r.t. functions of endomembrane system.
(1) Carbohydrate synthesis
(2) Lipoprotein synthesis
(3) Digestion of carbohydrates and nucleic acids
(4) Transport of modified substances

104. Observe the below given diagram and choose the **correct** option w.r.t. the labels.

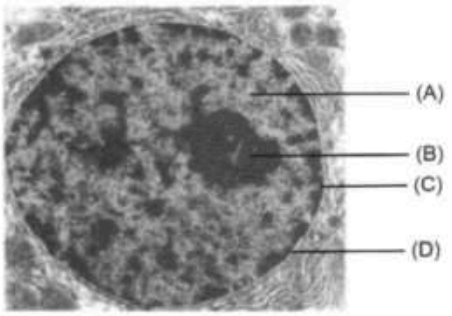


(a) Endoplasmic reticulum plays a role in the origin of 'E' and 'D'
(b) 'E' and 'A' help in the intracellular transport of nutrients and inorganic ions
(c) 'G', 'B' and 'C' contains nucleic acids
(d) Both 'C' and 'D' play no role in metabolism of cell in terms of degrading the materials
(e) 'B' and 'G' are found to be absent in prokaryotic cells

Choose the **correct** one(s):
(1) (a), (c) and (e) only
(2) (b) only
(3) (d) and (e) only
(4) (a) and (e) only

Space for Rough Work

105. Observe the following structure given below and choose the most appropriate option related to it.



(1) [redacted] consists of two parallel membranes with space of 10 to 50 mm, forming a barrier between cytoplasm and semi-fluid colloidal substance that fills the structure

(2) 'B' structure is not bound by membrane and representing the nucleoid region

(3) 'C' represents the passages through which movement of RNA and protein molecules occur in one direction, usually from outside to inside

(4) [redacted] structure is larger and numerous in cells which are actively carrying out protein synthesis

106. Read the given statements about the organelles performing specific functions in cell.

(P) All double membrane bound structures in eukaryotic cell contains dsDNA and 70S ribosomes

(Q) Mitochondria, [redacted] and chloroplast are semi-autonomous organelles

(R) All [redacted] cell organelles possess ATP, sugar and protein synthesising machinery

(S) Nucleus play a role in heredity

Based on the statements, pick the **correct** option.

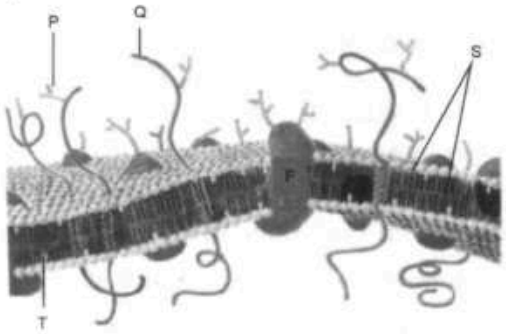
(1) P and Q are correct statements

(2) Q and R are correct statements

(3) S is the correct statement but P, Q and R are incorrect

(4) Q, R and S are correct statements but P is incorrect

107. Observe the given diagram and choose the **correct** option related to it.



(1) Both Q and [redacted] are integral proteins that can be removed easily

(2) S, T and F are able to perform flip-flop movement

(3) Q represents a protein molecule that is glycosylated

(4) [redacted] is representing the biomolecule that enables lateral movement of proteins, P and T are synthesised by ER

108. Read the given statements and choose the **incorrect** one.

(1) Change of colour from green to reddish during the ripening of tomato and chilli is due to transformation of chloroplasts to chromoplasts.

(2) Flagella of bacteria and eukaryotes have [redacted].

(3) The tubules of ER can be branched or unbranched structures and are common in cells involved in steroid synthesis.

(4) Two ribosomal subunits remain united with each other due to a specific concentration of [redacted] ions.

Space for Rough Work

109. Read the given Assertion (A) and Reason (R) statements and choose the **correct** option.

Assertion (A): The organelle which gives a cartwheel appearance is surrounded by massules.

Reason (R): Centrosome is a non-membrane bound organelle.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

110. Match **column-I** with **column-II**.

	Column-I		Column-II
a.	Mitochondria	(i)	Gives rise to SER
b.	Chloroplast	(ii)	Ribosomal RNA synthesis
c.	[redacted]	(iii)	Oxidative phosphorylation
d.	[redacted]	(iv)	Storage of starch

Choose the **correct** option.

(1) a(iii), b(ii), c(iv), d(i)

(2) a(i), b(ii), c(iv), d(iii)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(ii), b(iv), c(iii), d(i)

111. Identify the **correct** statement.

(1) Svedberg's unit is a direct measure of density and size of ribosomal subunits

(2) Varied number of cisternae are present in a Golgi complex

(3) Prokaryotic ribosomes are about 15 nm by 20 nm in size and are always found associated with mesosomes

(4) The cells of human cheek and onion peel have same outer boundary as the delimiting structure

112. Read the given statements and choose the **correct** option.

Statement (A): SER is associated with muscle contraction by release and uptake of Ca^{2+} ions as well as synthesis of glycoproteins.

Statement (B): Cell wall wards off the attacks of pathogens like viruses, bacteria and fungi and its composition does not vary among eukaryotic organisms as such.

(1) Both statements (A) and (B) are true

(2) Both statements (A) and (B) are false

(3) Only statement (A) is true

(4) Only statement (B) is true

113. Read the given Assertion (A) and Reason (R) statements and choose the **correct** option.

Assertion (A): Materials to be packaged in the form of vesicles from ER fuse with the concave face of golgi apparatus and move towards the convex face.

Reason (R): Golgi apparatus remains in close association with endoplasmic reticulum.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation (A)

(3) (A) is false but (R) is true

(4) Both (A) and (R) are false

114. The [redacted] differs from [redacted] as

(1) In the former 9 radial spokes and peripheral fibrils of tubulin proteins are present

(2) In the latter, the peripheral doublets of microtubule are connected to each other via linker protein

(3) The former is surrounded by amorphous pericentriolar material

(4) In the latter, total 27 microtubules are present

Space for Rough Work

115. Which of the following is the **correct** option w.r.t. chloroplast?
(1) Chlorophyll pigments are present in the space between inner and outer membrane
(2) Majority of these are found in mesophyll and BGA
(3) These can be of variable length (5 - 10 μm) and width (2 - 4 μm) but show spherical and discoid shape only
(4) The inner membrane of chloroplast is relatively less permeable than outer membrane

116. Both [redacted] and [redacted] are
(1) Surrounded by single membrane called tonoplast
(2) Rich in acidic hydrolases and play important role in cellular digestion
(3) Found in cytoplasm and occupy up to 90 percent of cell volume
(4) Sac like structures filled with digestive / excretory substances

117. Select the **incorrect** statement regarding plasma membrane.
(1) It is selectively permeable to some molecules present on either side of it
(2) Functioning of Na^+/K^+ pump is based on [redacted] transport
(3) The non-polar tail of phospholipids is made of [redacted] hydrocarbons
(4) It contains carbohydrate and cholesterol

118. (i) __A__ is the main arena of cellular activities in both plant and animal cells.
(ii) __B__ is used to monitor bacterial transformation with foreign DNA.

Choose the option that **correctly** fill in the blanks for **A** and **B**.

	A	B
(1)	Protoplasm	Pili
(2)	Nucleoplasm	Plastidome
(3)	Cytoplasm	Plasmid DNA
(4)	Stroma	Genomic DNA

119. Select the **odd** one out w.r.t. [redacted].
(1) Glycogen granules
(2) Polysome
(3) Phosphate granules
(4) Cyanophycean granules

120. Which of the following structures is responsible for increasing the surface area for their respective function and also found in [redacted]?
(1) Mesosome (2) Cilia
(3) Cristae (4) Cisternae

121. In potato tuber, the starch containing plastids are known as
(1) Aleuroplasts (2) Amyloplasts
(3) Chromoplasts (4) Elaioplasts

122. In chromosome, on the sides of primary constriction, disc shaped structures are present that are known as
(1) Centromere
(2) Kinetochore
(3) Diplosome
(4) Satellite

123. The structure which seals the ends of chromosomes and prevent their shortening is called
(1) Satellite (2) [redacted]
(3) [redacted] (4) Centrosphere

Space for Rough Work

124. The structure that forms the living component in the dead wall or form symplastic system between two cells, is
(1) Nucleoplasm (2) Middle lamella
(3) Plasmodesmata (4) Kinoplasm

125. Which of the following provides the precursors of enzymes for the formation of lysosomes?
(1) Mitochondria (2) Centrosome
(3) RER (4) SER

126. In mitochondria, the knob like protuberance called oxysome is present on/in
(1) Perimitochondrial space
(2) Outer membrane
(3) Cristae
(4) Cisternae

127. _____ forms the basal body of cilia that facilitate locomotion.
Choose the **correct** option to fill in the above blank.
(1) Ribosomes
(2) Endoplasmic reticulum
(3) Centrioles
(4) Centromere

128. The single membrane bound organelle that exhibits polymorphism on the basis of morphology, contents and functions is
(1) Golgi bodies
(2) Ribosomes
(3) Endoplasmic reticulum
(4) Lysosome

129. The structure that detaches from the maturing phase of the organelle 'X' may function as lysosome. This organelle 'X' is
(1) Vacuole
(2) Rough endoplasmic reticulum
(3) Golgi apparatus
(4) Smooth endoplasmic reticulum

130. Select the **odd** one out w.r.t. [redacted] subunit of ribosome.
(1) 18S rRNA (2) 28S rRNA
(3) 5.8S rRNA (4) 5S rRNA

131. Which of the following structures in prokaryotes can perform the function similar to the [redacted]?
(1) Polysome (2) Mesosomes
(3) Fimbriae (4) Chromatophores

132. The organelle, which is stained by vital stain 'Janus Green' and show great degree of variability in their shape, size and number is known as
(1) Brain of the cell
(2) Protein synthesis machinery of the cell
(3) Powerhouse of the cell
(4) Centrosome

133. Endomembrane system does **not** include
(1) SER (2) Centrosome
(3) Golgi complex (4) RER

134. Gas vacuoles are
(1) Responsible to provide buoyancy in protists
(2) Surrounded by a unit membrane
(3) Found in purple photosynthetic bacteria
(4) Identical in structure to that of sap vacuole

135. The [redacted] cell is the
(1) Red blood cell (2) Nerve cell
(3) Mesophyll cell (4) Ostrich egg

Space for Rough Work

[ZOOLOGY]

136. Which of the following cells make up more than one-half the volume of the neural tissue in our body?
(1) Neurons (2) Neuroglial cells
(3) Chondrocytes (4) Fibroblasts

137. Arrange the following in increasing order of their structural complexity.
a. Nucleotide b. Nitrogenous base
c. Nucleic acid d. Nucleoside
Choose the **correct** option.
(1) b → d → a → c (2) d → b → a → c
(3) a → b → c → d (4) c → a → d → b

138. A tissue shows the following features:
- A free surface
- A little intercellular matrix
- A basement membrane

Identify the type of tissue and choose the option that cannot be **true** for it.
(1) It may be stratified to withstand mechanical stresses
(2) It shows regeneration capacity
(3) It assists in secretion
(4) It contains only cube-shaped cells

139. Choose the **incorrect** match.

(1)	Areolar tissue	–	Serves as a support framework for epithelium
(2)	Adipose tissue	–	Stores excess of nutrients which are not used immediately
(3)	Dense regular connective tissue	–	Specialized connective tissue
(4)	Blood	–	Transports gases

140. Which of the following features is true w.r.t. muscle fibres present in the wall of stomach?
(1) They are involuntary
(2) Cell junctions are absent in them
(3) They are multi-nucleated
(4) They are striated

141. Consider the following features:
(a) Produce histamine
(b) Produce heparin
(c) Found in blood
(d) Absent in loose connective tissues
How many of the above mentioned features is/are true for mast cells?
(1) One
(2) Three
(3) Four
(4) Two

142. Arrival of the disturbance at the _____ triggers events that may cause _____ of the adjacent neurons and other cells.
Read the following:
(A) Output zone; Inhibition
(B) Neuron's ending; Stimulation
(C) Dendrite; Only excitation
Which of the above would correctly fill the respective blanks?
(1) (A) only
(2) (B) only
(3) (A) and (B) only
(4) (C) only

Space for Rough Work

143. **Assertion (A):** Simple squamous epithelium is ideal for diffusion and filtration of substances.

Reason (R): It is composed of a single thin layer of flattened cells resting on a cellular basement membrane.

In the light of above statements, choose the **correct** option.

(1) Both (A) and (R) are true, and (R) explains (A) correctly

(2) Both (A) and (R) are true, but (R) does not explain (A) correctly

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

144. In a hypothetical condition, if a thermolabile enzyme is exposed to prolonged heating at nearly 195°C in a lab, then, what will be the most possible outcome?

(1) The enzyme will still function normally

(2) The high temperature will affect the tertiary structure of the enzyme, so it will not work effectively now

(3) Upon providing normal temperature conditions, the enzyme will regain its function

(4) The glycosidic bonds present between the amino acids of the enzymes will get broken, rendering it to become ineffective

145. How many cells mentioned in the box given below is/are absent in the ground substance of loose connective tissues?

Macrophage, Plasma cell, Fibroblast, Erythrocyte, Osteocyte, Adipocyte

Choose the **correct** option.

(1) One

(2) Two

(3) Three

(4) Four

146. Match List I with List II w.r.t. humans.

	List I		List II
a.	Isolated glandular cells	(i)	Digestive enzymes like lipases
b.	Exocrine secretion	(ii)	Hormone
c.	Compound epithelium	(iii)	Goblet cells secreting mucus
d.	Endocrine secretion	(iv)	Moist surface of buccal cavity

Choose the **correct** option.

(1) a(i), b(ii), c(iii), d(iv) (2) a(iii), b(i), c(iv), d(ii)

(3) a(iii), b(iv), c(ii), d(i) (4) a(i), b(iv), c(iii), d(ii)

147. Consider the following statements regarding epithelial tissue:

(i) The nuclei of cells in columnar epithelium is present near the base of cells.

(ii) Simple cuboidal epithelium is always found in areas subjected to mechanical stress such as wear and tear.

(iii) Cuboidal epithelium present in the kidney tubule plays an important role in secretion and reabsorption.

(iv) Ciliated epithelium is primarily involved in absorption as cilia increases surface area.

Which of the above statements are **correct**?

(1) (i) and (ii) only (2) (ii) and (iii) only

(3) (i) and (iii) only (4) (i), (ii), (iii) and (iv)

148. Select the option that correctly represents the difference between adhering and tight junctions.

(1) Latter help in cell communication by connecting the cytoplasm of adjoining cells

(2) Former perform cementing to keep the neighbouring cells together

(3) Latter help in rapid transfer of only small ions

(4) Former help to stop substances from leaking across a tissue

Space for Rough Work

149. The structure responsible for attachment of bone to bone
(1) Is a type of loose connective tissue
(2) Belongs to the same category to which tendon belongs to
(3) Lacks fibroblasts
(4) Is the site for RBC production

150. In humans, the cuboidal epithelium does not line the
(1) Ducts of glands
(2) Thyroid follicles
(3) Ovary
(4) Lining of intestine and stomach

151. Complete the analogy by selecting the **correct** option w.r.t. humans.
Stratified squamous epithelium : Dry surface of skin :: _____ : Inner lining of pancreatic duct
(1) Compound epithelium
(2) Simple squamous epithelium
(3) Ciliated columnar epithelium
(4) Keratinised cuboidal epithelium

152. The distinguishing feature between bone and cartilage is that
(1) In former, cells are present within the lacunae.
(2) Latter interacts with skeletal muscles to bring about movements.
(3) Former has concentric rings in its matrix and is the main site for erythropoiesis in adults.
(4) Latter has hard and non-pliable ground substance.

153. Read the following statements:
Statement (A): Some secondary metabolites have ecological importance.
Statement (B): Lecithin contains only polar groups in its structure.
Choose the **correct** option.
(1) Both statements (A) and (B) are true
(2) Both statements (A) and (B) are false
(3) Only statement (A) is true
(4) Only statement (B) is true

154. 'X' epithelium lines the inner surface of the fallopian tube, while 'Y' muscle forms the wall of fallopian tube. Select the correct option w.r.t. 'X' and 'Y'.
(1) 'X' helps in movement of ova through the oviduct
(2) 'X' and 'Y' share the common origin as both are derived from the ectodermal layer.
(3) 'Y' is voluntary in nature.
(4) 'X' and 'Y' possess only microvilli on their free surface, and secrete nutrients required for the development of ova.

155. How many features mentioned in the box given below is/are not common between the muscle fibres present in biceps and in wall of heart?

Cylindrical shape, Uninucleated condition, Striated appearance, Presence of intercalated discs, Presence of gap junctions, Branching

Select the **correct** option.
(1) Two
(2) One
(3) Four
(4) Three

156. Division of labour at cellular level is seen in all of the following, **except**
(1) *Hydra*
(2) *Paramecium*
(3) Pigeon
(4) Camel

157. The inner lining of hollow structures like bronchioles is lined by
(1) Brush-bordered cuboidal epithelium
(2) Squamous epithelium
(3) Ciliated epithelium
(4) Compound epithelium

Space for Rough Work

158. Choose the **correct** match.

(1)	Glycogen	–	Storage polysaccharide in animals
(2)	Maltose	–	Reducing sugar composed of fructose
(3)	Cellulose	–	Easily digested by humans
(4)	Sucrose	–	Reducing sugar due to free aldehyde group

159. Read the following statements w.r.t. proteins and amino acids:

A. All naturally occurring amino acids found in proteins are non-essential amino acids.

B. Peptide bond formation involves the carboxyl group of one amino acid and amino group of another.

C. A protein thread does not exist throughout as an extended rigid rod.

D. Denaturation of protein results in the loss of only the primary structure of the protein.

Which of the following is/are **correct**?

(1) Only B and C

(2) Only C

(3) Only A, B, and C

(4) Only A, C and D

160. Which of the following functions is associated with neuroglial cells?

(1) Reception of stimuli

(2) Protection and support

(3) Generation of impulse

(4) Transmission of nerve impulse

161. Which of the following differentiates lipids from both polysaccharides and proteins?

(1) Found in acid-soluble pool (obtained after chemical analysis of a living tissue)

(2) Not polymeric

(3) Presence of carbon in their structure

(4) Presence of covalent bonds in their structure

162. Match List I with List II w.r.t. secondary metabolites.

	List I		**List II**
a.	Polymeric substance	(i)	Concanavalin A
b.	Drug	(ii)	Rubber
c.	Alkaloid	(iii)	Curcumin
d.	Lectin	(iv)	Codeine
		(v)	Cellulose

Choose the **correct** option.

(1) a(i), b(iii), c(ii), d(iv)

(2) a(ii), b(i), c(iii), d(v)

(3) a(ii), b(iii), c(iv), d(i)

(4) a(iv), b(i), c(ii), d(iii)

163. Which of the following options best explains as to why enzymes are highly specific?

(1) Enzymes are made up of proteins which are homopolymers

(2) Active sites of enzymes have a particular shape and specific amino acid residues

(3) Enzymes increase the activation energy required for reaction to proceed

(4) Enzymes are reusable

Space for Rough Work

164. Digestive enzyme that breaks down dietary proteins into smaller peptides in the stomach, belongs to which class of enzymes?
(1) Class IV
(2) Class V
(3) Class III
(4) Class I

165. The nitrogenous base that is absent in RNA but present in the structure of DNA _____.
Choose the option to fill in the blank correctly.
(1) Pairs exclusively with cytosine
(2) Is a purine nitrogenous base
(3) Is also called 5′ methyl uracil that pairs with adenine *via* two hydrogen bonds
(4) Is made up of two heterocyclic rings

166. 'X' constitutes 10-15% of the total cellular mass. 'X' may contain
(a) Disulphide bond (b) Hydrogen bond
(c) Peptide bond
Choose the option only with the **correct** set(s).
(1) (a), (b) and (c) only
(2) (c) only
(3) (b) and (c) only
(4) (a) and (b) only

167. Read the given statements:
Statement (A): GLUT-4 is a transporter protein responsible for transporting a hexose sugar into the cells.
Statement (B): The one feature common to all those compounds found in the acid-soluble pool is that they have molecular weights ranging from 18 to around 800 daltons approximately.
Choose the **correct** option:
(1) Both statements (A) and (B) are incorrect
(2) Both statements (A) and (B) are correct
(3) Only statement (A) is correct
(4) Only statement (B) is correct

168. If we analyze the chemical composition of a living tissue of human body, from abundance point of view and arrange them in correct chronological order, we can say that _____ is the most abundant element, whereas _____ is the least found element.
Choose the **correct** option to fill the respective blanks.
(1) Sulphur; Calcium
(2) Oxygen; Silicon
(3) Magnesium; Hydrogen
(4) Carbon; Nitrogen

169. Choose the **incorrect** match.

(1)	Catalase	–	Haem is the prosthetic group
(2)	NADP	–	Co-enzyme containing vitamin niacin
(3)	Peroxidase	–	Breaks hydrogen peroxide into water and oxygen
(4)	Carboxypeptidase	–	Remains active even after zinc is removed from it

170. If a double stranded DNA molecule has 30% adenine, then, the percentage of cytosine present in this DNA molecule will be
(1) 20 (2) 30
(3) 70 (4) 40

Space for Rough Work

171. **Assertion (A):** Arachidonic acid (16-carbon) has a higher melting point in comparison to that of palmitic acid (20-carbon).

Reason (R): Palmitic acid is an unsaturated fatty acid with 20-carbon chain while, arachidonic acid is a 16-carbon containing unsaturated fatty acid.

In the light of above statements, choose the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true, (R) is false

(4) Both (A) and (R) are false

172. Choose the **incorrect** statement.

(1) Aquaporins are proteins.

(2) Cholesterol is a derived lipid containing five fused carbon rings.

(3) Inorganic catalysts function efficiently at high temperature and pressure.

(4) Physiological reactions are enzyme catalyzed reactions.

173. For the formation of a linear polysaccharide chain made up of 50 glucose units, 'Q' bonds are formed and 'P' water molecules are formed and eliminated. Choose the **correct** option w.r.t. 'P' and 'Q', respectively.

(1) 50; 50 (2) 49; 49

(3) 50; 49 (4) 49; 50

174. If after addition of an inhibitor in an enzyme catalysed reaction, the enzyme shows increased Km value to achieve same V_{max} as was present before addition of inhibitor, then the inhibitor is most likely

(1) A non-competitive inhibitor that binds at a site other than the active site.

(2) A competitive inhibitor that resembles the substrate molecule in structure and binds to the active site of the enzyme.

(3) A reversible inhibitor that denatures the enzymes by increasing the activation energy of the reaction.

(4) An irreversible inhibitor that destroys the enzyme's active site.

175. Chitin is best described as a

(1) Homopolymer of glucose with phosphate group found in the cell wall of bacteria

(2) Heteropolymer of glucose containing nitrogen derivative

(3) Storage polysaccharide in plants

(4) Homopolysaccharide that is formed of amino sugar known as N-acetylglucosamine

176. Consider the following sequence of events during protein organisation:

A. Formation of α-helix and β-pleated sheet

B. Folding of a long polypeptide chain, like a hollow woolen ball

C. Association of two or more polypeptide chains/subunits

Which level of protein structure is represented by event 'C' and which bond/interaction stabilizes it?

(1) Tertiary structure – Hydrogen bond

(2) Secondary structure – Peptide bond

(3) Quaternary structure – Non-covalent interactions

(4) Primary structure – Disulphide bond

Space for Rough Work

177. Which among the following are not homopolymers?

A. Antibody B. Starch
C. Cellulose D. Collagen
E. RuBisCO

Choose the **correct** option.

(1) A and B only (2) A, D and E only
(3) B, C and D only (4) B and C only

178. Select the **correct** statement.

(1) The heterocyclic moiety found in nucleic acids are the nitrogenous bases only.
(2) The sugar found in polynucleotides is always a monosaccharide hexose sugar.
(3) Cellulose does not contain complex helices as found in starch, and hence, it cannot hold iodine molecules.
(4) Cellulose is a polysaccharide consisting of glucose and galactose as its monomers.

179. A non-digestible carbohydrate in humans is inulin, which is

(1) A polymer of a non-reducing sugar.
(2) A polymer whose monomeric units are joined together *via* peptide bonds
(3) A polymer of hexose sugar, called fructose
(4) Not found in plants, but is present in fungi

180. Ester bonds present in deoxyribonucleic acid, form the backbone connecting

(1) Only two phosphate groups together
(2) Phosphate only with nitrogenous bases
(3) Ribose sugar with nitrogenous bases
(4) 5′ carbon of one deoxyribose sugar to the 3′ carbon of the next sugar, with a phosphate group acting as the bridge between them

❑ ❑ ❑

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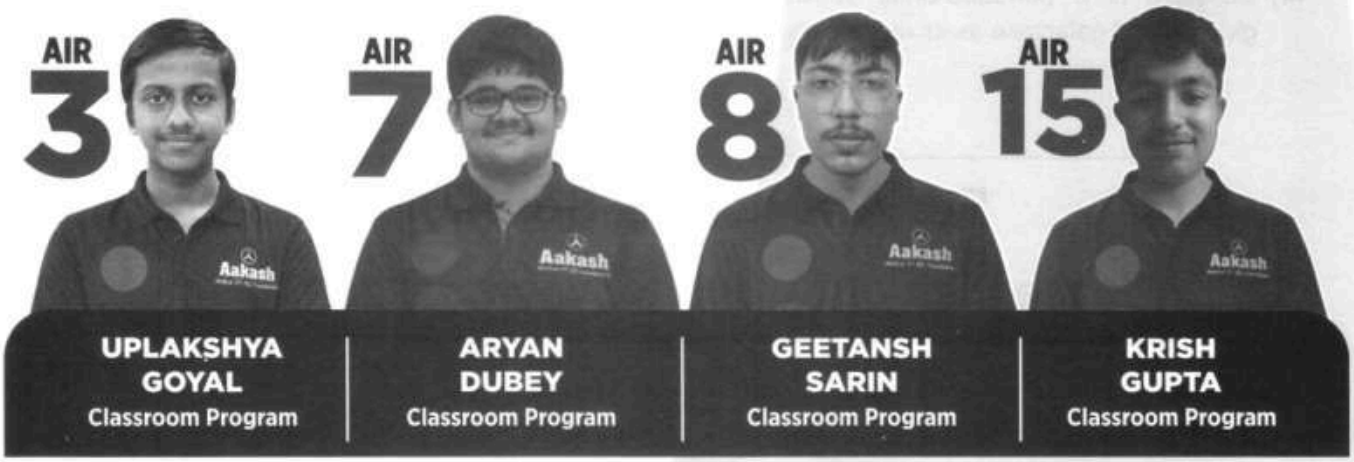
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ANSWERS

1. (3)	37. (2)	73. (1)	109. (2)	145. (2)
2. (2)	38. (3)	74. (2)	110. (3)	146. (2)
3. (4)	39. (1)	75. (4)	111. (2)	147. (3)
4. (1)	40. (4)	76. (4)	112. (2)	148. (2)
5. (3)	41. (4)	77. (2)	113. (3)	149. (2)
6. (2)	42. (3)	78. (3)	114. (4)	150. (4)
7. (4)	43. (1)	79. (2)	115. (4)	151. (1)
8. (3)	44. (1)	80. (4)	116. (4)	152. (3)
9. (3)	45. (3)	81. (3)	117. (2)	153. (3)
10. (3)	46. (2)	82. (1)	118. (3)	154. (1)
11. (1)	47. (4)	83. (3)	119. (2)	155. (3)
12. (3)	48. (1)	84. (4)	120. (1)	156. (2)
13. (3)	49. (2)	85. (1)	121. (2)	157. (3)
14. (3)	50. (2)	86. (4)	122. (2)	158. (1)
15. (3)	51. (3)	87. (2)	123. (3)	159. (1)
16. (2)	52. (4)	88. (1)	124. (3)	160. (2)
17. (4)	53. (3)	89. (1)	125. (3)	161. (2)
18. (4)	54. (3)	90. (2)	126. (3)	162. (3)
19. (2)	55. (2)	91. (3)	127. (3)	163. (2)
20. (2)	56. (1)	92. (2)	128. (4)	164. (3)
21. (2)	57. (1)	93. (4)	129. (3)	165. (3)
22. (1)	58. (2)	94. (4)	130. (1)	166. (1)
23. (4)	59. (4)	95. (4)	131. (2)	167. (2)
24. (1)	60. (1)	96. (3)	132. (3)	168. (2)
25. (4)	61. (1)	97. (2)	133. (2)	169. (4)
26. (3)	62. (1)	98. (1)	134. (3)	170. (1)
27. (1)	63. (2)	99. (3)	135. (4)	171. (4)
28. (4)	64. (1)	100. (4)	136. (2)	172. (2)
29. (1)	65. (4)	101. (1)	137. (1)	173. (2)
30. (3)	66. (1)	102. (2)	138. (4)	174. (2)
31. (3)	67. (2)	103. (1)	139. (3)	175. (4)
32. (4)	68. (1)	104. (4)	140. (1)	176. (3)
33. (1)	69. (1)	105. (4)	141. (4)	177. (2)
34. (4)	70. (1)	106. (3)	142. (3)	178. (3)
35. (4)	71. (2)	107. (3)	143. (3)	179. (3)
36. (1)	72. (1)	108. (4)	144. (2)	180. (4)